

Conversion between Array, List, and Dictionary in C#

In this article, we will discuss how to perform **Conversion Between Array List and Dictionary in C#**. Please read our previous article where we discussed [Dictionary in C#](#) with examples. As part of this article, we will discuss the following six things.

1. Convert an array to a List – Use [ToList\(\)](#) method
2. Convert a list to an array – Use [ToArray\(\)](#) method
3. Convert a List to a Dictionary – Use [ToDictionary\(\)](#) method
4. Convert an array to a Dictionary – Use [ToDictionary\(\)](#) method
5. Convert a Dictionary to an array – Use [ToArray\(\)](#) method on the [Values](#) Property of the dictionary object
6. Convert a Dictionary to a List – Use the [ToList\(\)](#) method on the [Values](#) Property of the dictionary object

Let us understand this with an example.

The code is self-explained. Please go through the comments.

```
using System;
using System.Collections.Generic;
using System.Linq;
namespace CollectionDemo
{
    public class Program
    {
        static void Main(string[] args)
        {
            //Create Employee object
            Employee emp1 = new Employee()
            {
                ID = 101,
                Name = "Pranaya",
                Gender = "Male",
                Salary = 20000
            };
            Employee emp2 = new Employee()
            {
                ID = 102,
                Name = "Priyanka",
                Gender = "Female",
                Salary = 30000
            };
            Employee emp3 = new Employee()
            {
                ID = 103,
                Name = "Anurag",
                Gender = "Male",
                Salary = 40000
            }
        }
    }
}
```

```

};
// Create an array of employees with size 3
// and then Store the 3 employees into the array
Employee[] arrayEmployees = new Employee[3];
arrayEmployees[0] = emp1;
arrayEmployees[1] = emp2;
arrayEmployees[2] = emp3;
// To convert an array to a List, use ToList() method
List<Employee> listEmployees = arrayEmployees.ToList();
foreach (Employee emp in listEmployees)
{
    Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                        emp.ID, emp.Name, emp.Gender, emp.Salary);
}
Console.WriteLine();
// To convert a List to an array, use ToArray() method
Employee[] arrayAllEmployeesFromList = listEmployees.ToArray();
foreach (Employee emp in arrayAllEmployeesFromList)
{
    Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                        emp.ID, emp.Name, emp.Gender, emp.Salary);
}
Console.WriteLine();
// To convert a List to a Dictionary, use ToDictionary() method
Dictionary<int, Employee> dictionaryEmployees =
listEmployees.ToDictionary(employee => employee.ID, employee => employee);
foreach (KeyValuePair<int, Employee> keyValuePairEmployees in
dictionaryEmployees)
{
    Console.WriteLine("Key = {0}", keyValuePairEmployees.Key);
    Employee emp = keyValuePairEmployees.Value;
    Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                        emp.ID, emp.Name, emp.Gender, emp.Salary);
}
Console.WriteLine();
// To convert an Array to a Dictionary, use ToDictionary() method
Dictionary<int, Employee> dictionaryEmployeesFromArray =
arrayEmployees.ToDictionary(employee => employee.ID, employee => employee);
// Loop thru the dictionary and print the key/value pairs
foreach (KeyValuePair<int, Employee> kvp in dictionaryEmployeesFr
{
    Console.WriteLine("Key = {0}", kvp.Key);
    Employee emp = kvp.Value;
    Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                        emp.ID, emp.Name, emp.Gender, emp.Salary);
}
// To Convert a dictionary to an array, use ToArray method on the
// Property of the dictionary object
Employee[] arrayAllEmployeesFromDictionary =
dictionaryEmployeesFromArray.Values.ToArray();
foreach (Employee emp in arrayAllEmployeesFromDictionary)

```

```

        {
            Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                               emp.ID, emp.Name, emp.Gender, emp.Salary);
        }
        Console.WriteLine();
        // To Convert a dictionary to a List, use To List method on the Va
        // Property of the dictionary object
        List<Employee> listAllEmployeesFromDictionary =
dictionaryEmployeesFromArray.Values.ToList();
        foreach (Employee emp in listAllEmployeesFromDictionary)
        {
            Console.WriteLine("ID = {0}, Name = {1}, Gender ={2}, Salary =
                               emp.ID, emp.Name, emp.Gender, emp.Salary);
        }
        Console.ReadKey();
    }
}
public class Employee
{
    public int ID { get; set; }
    public string Name { get; set; }
    public string Gender { get; set; }
    public int Salary { get; set; }
}
}

```

Output:

```
ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
ID = 103, Name = Anurag, Gender =Male, Salary = 40000

ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
ID = 103, Name = Anurag, Gender =Male, Salary = 40000

Key = 101
ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
Key = 102
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
Key = 103
ID = 103, Name = Anurag, Gender =Male, Salary = 40000

Key = 101
ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
Key = 102
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
Key = 103
ID = 103, Name = Anurag, Gender =Male, Salary = 40000
ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
ID = 103, Name = Anurag, Gender =Male, Salary = 40000

ID = 101, Name = Pranaya, Gender =Male, Salary = 20000
ID = 102, Name = Priyanka, Gender =Female, Salary = 30000
ID = 103, Name = Anurag, Gender =Male, Salary = 40000
```

In the next article, I am going to discuss [List vs Dictionary in C#](#) with examples. Here, in this article, I try to explain the **Conversion Between Array List and Dictionary in C#** with an example. I hope this article will help you with your need. I would like to have your feedback. Please post your feedback, question, or comments about this article.